Algorithm 1 Meta-evolution of symbolic-regression operators

Input Baseline bundle B_0; training tasks $\mathcal{T}$; fitness F; population target M; offspring count O; G generations, including G_s final simplification generations; initial and survivor seed counts $n_0 \le n_1$.

$\mathcal{P} \leftarrow \{B_0\} \cup \textsc{InitialExplore}(B_0, M-1)$
$\mathcal{A} \leftarrow \textsc{Evaluate}(\mathcal{P}, \mathcal{T}, F, n_0)$ # Store per-task, per-seed scores
for $g = 1, \ldots, G$ **do**
  $s \leftarrow [g > G - G_s]$ # Switch from performance to simplification
  if s and this is the first simplification generation **then**
    $\mathcal{P} \leftarrow \textsc{SeedFrontier}(\mathcal{A}, M)$ # Recover the archive's score–LOC tradeoffs
  Launch survivor top-ups to n_1 seeds; $\mathcal{C} \leftarrow \varnothing$
  for j in a shuffled schedule of O slots, balanced across operator types **do**
    $B \leftarrow \textsc{Tournament}(\mathcal{P}, 2)$ # Higher mean fitness wins
    $m \leftarrow$ `simplify` if s, otherwise a uniform feasible meta-mutation mode
    $U \leftarrow \textsc{SourceOperators}(\mathcal{P}, B, j, m)$
    $q \leftarrow \textsc{Prompt}(j, m, U, \textsc{OptionalTraceFeedback}(B, \mathcal{T}))$
    $u \leftarrow \textsc{ValidatedLLMCompletion}(q)$ # Sample ensemble; retry invalid code
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{B[j \leftarrow u]\}$ # Replace one slot; inherit the rest
  Evaluate $\mathcal{C}$ on n_0 seeds, overlapping survivor top-ups
  Merge new observations into $\mathcal{A}$; update task means and overall fitness
  $\mathcal{P} \leftarrow \begin{cases} \textsc{ScoreLOCSurvivors}(\mathcal{P} \cup \mathcal{C}, M), & s, \\ \textsc{TaskDiverseSurvivors}(\mathcal{P} \cup \mathcal{C}, M), & \text{otherwise} \end{cases}$
Top up the last survivors to n_1 seeds
$\mathcal{Q} \leftarrow \textsc{Finalists}(\mathcal{A}, \mathcal{P})$; evaluate $\mathcal{Q}$ on fresh shared seeds
return fresh scores and $\arg\max_{B \in \mathcal{Q}} \widehat{F}_{\text{fresh}}(B)$

Modes. Explore, refine, crossover, simplify. Refine/simplify use the selected bundle's operator when available; crossover samples two source operators uniformly. Optional trace feedback concerns unsolved tasks. Failed or infeasible proposals use the implementation's bounded retries and fallbacks.

Survivors. Task diversity retains distinct best positive-rate solvers, then backfills to at least M by overall fitness. Score–LOC selection takes a best candidate per LOC bucket, Pareto-filters these winners, then backfills to M by fitness. The archive mean is $\widehat{F}(B) = |\mathcal{T}|^{-1} \sum_{t \in \mathcal{T}} n_B^{-1} \sum_{r=1}^{n_B} F(B, t, r)$.

Scope. The task-diverse, population-reevaluation variant is shown. Finalists may be the terminal population or an archive shortlist; the current identification pass uses the top K code-distinct archive bundles by shrinkage-adjusted training score. Held-out validation and benchmark evaluation are separate.